COMPUTER PROGRAMING LAB REPORT # 3

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SECTION	"A"
SEMISTER	2ND
SUBJECT	COMPUTER PROGRAMMING

1. *Objective*

The objective of this lab is to master decision-making in C++ by implementing:

- Relational Operators for value comparison.
- Logical Operators for combining multiple conditions.
- Decision Control Structures including if, if-else, and else-if cascades.

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2. *Technical Theory*

2.1 Relational Operators

Used to compare two values. The result is always a boolean (true or false).

- < (Less than), > (Greater than)
- <= (Less than or equal to), >= (Greater than or equal to)
- == (Equal to), != (Not equal to)
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2.2 *Logical Operators*

Used to form complex conditions by combining relational expressions.

- && (Logical AND): True if both conditions are true.
- || (Logical OR): True if at least one condition is true.
- ! (Logical NOT): Reverses the logical state of its operand.
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2.3 *Decision Control Structures*

By default, C++ code executes sequentially. Decision structures allow the program to skip or execute specific blocks based on conditions.

- if: Executes a block only if the condition is true.
- if-else: Provides an alternative block if the condition is false.
- else-if: Allows for multiple independent conditions to be checked in sequence.

Task 1:

Basic if Statement:

Problem: Take a number as input. If it is greater than 100, print "The number is greater than hundred".

```
#include <iostream>
using namespace std;

int main() {
    int num;
    cout << "Enter a number: ";
    cin >> num;

    if (num > 100) {
        cout << "The number is greater than hundred" << endl;
    }
    return 0;
}
```

Task 2:

The if-else Statement

Problem: Take two numbers as input and subtract the larger number from the smaller one (or find the absolute difference)

```cpp
#include <iostream>
using namespace std;

int main() {
    int a, b, result;
    cout << "Enter two numbers: ";
    cin >> a >> b;

    if (a < b) {
        result = a - b;
        cout << "Subtracting larger from smaller: " << result << endl;
    } else {
        result = b - a;
        cout << "Subtracting larger from smaller: " << result << endl;
    }
    return 0;
}
```

Task 3:

The else-if Statement (Multi-way Decision)

Problem: Implement a grade or category check based on multiple ranges.

```cpp
#include <iostream>
using namespace std;

int main() {
    int score;
    cout << "Enter score: ";
    cin >> score;

    if (score >= 90) {
        cout << "Grade: A" << endl;
    } else if (score >= 80) {
        cout << "Grade: B" << endl;
    } else if (score >= 70) {
        cout << "Grade: C" << endl;
    } else {
        cout << "Grade: F" << endl;
    }
    return 0;
}
```

4. Conclusion

In this lab, we successfully implemented decision-making logic in C++. We observed that ifelse structures provide a way to handle binary choices, while the else-if ladder is essential for handling multiple mutually exclusive conditions efficiently.